

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2021 P1H Worked Solutions

May 2021 | P1H

33 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Standard & Compound Units**

May 2021 P1H - May 2021 | P1H

- 1** A plane flew from Madrid to Dubai.

The distance the plane flew was 5658 km.

The flight time was 8 hours 12 minutes.

Work out the average speed of the plane.

..... km/h

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 1** **Marks: 3****TOPIC: Standard & Compound Units**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$8 \text{ h } 12 \text{ min} = 8 + \frac{12}{60} = 8.2 \text{ hours}$$

$$\text{average speed} = \frac{5658}{8.2} = 690$$

FINAL ANSWER

690 km/h.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 2**TOPIC: **Sequences**

May 2021 P1H - May 2021 | P1H

- 2 Here are the first 4 terms of an arithmetic sequence.

85 79 73 67

Find an expression, in terms of n , for the n th term of the sequence.

(Total for Question 2 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 2****Marks: 2****TOPIC: Sequences**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The sequence is**

The sequence is

$$85, 79, 73, 67$$

2. The common difference isThe common difference is -6 .

$$n\text{th term} = 85 + (n - 1)(-6)$$

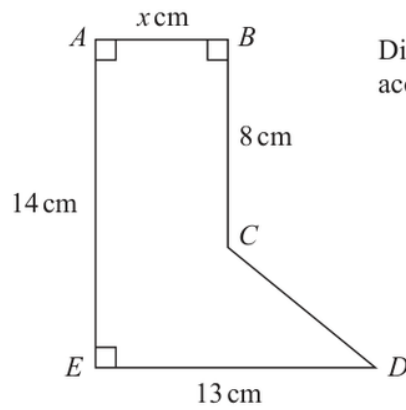
$$= 85 - 6n + 6 = 91 - 6n$$

FINAL ANSWER

$$91 - 6n.$$

PAST PAPER QUESTION**Q#: 3****Marks: 4**TOPIC: **Area & Perimeter**

May 2021 P1H - May 2021 | P1H

3Diagram **NOT**
accurately drawnThe diagram shows the shape $ABCDE$.The area of the shape is 91.8 cm^2 Work out the value of x . $x = \dots\dots\dots$ **(Total for Question 3 is 4 marks)**

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC: Area & Perimeter**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the shape into a**

Split the shape into a rectangle and a triangle.

2. Rectangle area

Rectangle area:

$$14x$$

3. Use trigonometryThe triangle has base $13 - x$ and height 6, so its area is

$$\frac{1}{2}(13 - x)(6) = 39 - 3x$$

4. Total area

Total area:

$$14x + 39 - 3x = 91.5$$

$$11x = 52.5$$

$$x = 4.772 \dots$$

FINAL ANSWER

4.77 cm.

PAST PAPER QUESTION**Q#: 4****Marks: 3**TOPIC: **Ratio Toolkit**

May 2021 P1H - May 2021 | P1H

- 4 On a farm there are chickens, ducks and pigs.

The ratio of the number of chickens to the number of ducks is $7:2$

The ratio of the number of ducks to the number of pigs is $5:9$

There are 36 pigs on the farm.

Work out the number of chickens on the farm.

(Total for Question 4 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 4

Marks: 3

TOPIC: **Ratio Toolkit**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Ducks pigs**

Ducks : pigs = 5 : 9.

2. Split the ratio

There are 36 pigs, so

$$9 \text{ parts} = 36$$

$$1 \text{ part} = 4$$

$$\text{ducks} = 5 \times 4 = 20$$

3. Chickens ducks

Chickens : ducks = 7 : 2.

$$2 \text{ parts} = 20$$

$$1 \text{ part} = 10$$

$$\text{chickens} = 7 \times 10 = 70$$

FINAL ANSWER

70

PAST PAPER QUESTION**Q#: 5****Marks: 6****TOPIC: Rearranging Formulas**

May 2021 P1H - May 2021 | P1H

- 5 (a) Expand and simplify $3x(2x + 3) - x(3x + 5)$

.....
(2)

- (b) Make t the subject of the formula $p = at - d$

.....
(2)

Given that $\frac{w^5 \times w^n}{w^3} = w^{10}$

- (c) work out the value of n .

$n =$
(2)

(Total for Question 5 is 6 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 6****TOPIC: Rearranging Formulas**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\begin{aligned}
 3x(2x + 3) - x(3x + 5) &= 6x^2 + 9x - 3x^2 - 5x \\
 &= 3x^2 + 4x
 \end{aligned}$$

2. (b)

(b)

$$p = at - d$$

$$p + d = at$$

$$t = \frac{p + d}{a}$$

3. (c)

(c)

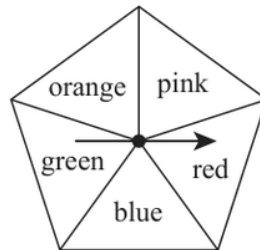
$$\frac{w^5 \times w^n}{w^3} = w^{n+2} = w^{10}$$

4. Rearrange formulaSo $n + 2 = 10$, hence $n = 8$.**FINAL ANSWER**(a) $3x^2 + 4x$, (b) $t = \frac{p+d}{a}$, (c) $n = 8$

PAST PAPER QUESTION**Q#: 6****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P1H - May 2021 | P1H

- 6 Grace has a biased 5-sided spinner.



Grace is going to spin the arrow on the spinner once.

The table below gives the probabilities that the spinner will land on red or on blue or on green.

Colour	Red	Blue	Green	Orange	Pink
Probability	0.20	0.12	0.08		

The probability that the spinner will land on orange is 3 times the probability that the spinner will land on pink.

- (a) Work out the probability that the spinner will land on orange.

.....
(3)

Grace spins the arrow on the spinner 150 times.

- (b) Work out an estimate for the number of times the spinner lands on blue.

.....
(2)

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the probability of pink**

Let the probability of pink be x . Then the probability of orange is $3x$.

$$0.20 + 0.12 + 0.08 + x + 3x = 1$$

$$4x = 0.60$$

$$x = 0.15$$

2. Calculate probability

So the probability of orange is

$$3x = 0.45$$

3. Expected blue sweets

Expected blue sweets:

$$150 \times 0.12 = 18$$

FINAL ANSWER

pink 0.15, orange 0.45, blue 18.

PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Solving Inequalities**

May 2021 P1H - May 2021 | P1H

7 $-4 \leq 2y < 6$

y is an integer.

(a) Write down all the possible values of y .

.....
(2)

(b) Solve the inequality $7t - 3 \leq 2t + 31$

Show your working clearly.

.....
(2)

.....
(Total for Question 7 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 7****Marks: 4****TOPIC: Solving Inequalities**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$-4 \leq 2y < 6$$

2. Divide by 2

Divide by 2:

$$-2 \leq y < 3$$

3. Calculate valueSince y is an integer,

$$y = -2, -1, 0, 1, 2$$

4. (b)

(b)

$$7t - 3 \leq 2t + 31$$

$$5t \leq 34$$

$$t \leq \frac{34}{5}$$

FINAL ANSWER

$$y = -2, -1, 0, 1, 2, \text{ and } t \leq \frac{34}{5}.$$

PAST PAPER QUESTION**Q#: 8****Marks: 4****TOPIC: Powers, Roots & Standard Form**

May 2021 P1H - May 2021 | P1H

- 8 The table shows the populations of five countries.

Country	Population
China	1.4×10^9
Germany	8.2×10^7
Sweden	9.9×10^6
Fiji	9.1×10^5
Malta	4.3×10^5

- (a) Work out the difference between the population of China and the population of Germany.
Give your answer in standard form.

.....
(2)

Given that

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

- (b) work out the value of k .
Give your answer correct to the nearest whole number.

 $k =$
(2)
(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 4****TOPIC: Powers, Roots & Standard Form**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

For part (a),

$$1.4 \times 10^9 = 1400000000$$

$$8.2 \times 10^7 = 82000000$$

2. Difference

Difference:

$$1400000000 - 82000000 = 1318000000$$

$$1318000000 = 1.318 \times 10^9$$

3. Convert standard form

For part (b),

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

4. Convert standard form

So

$$k = \frac{\text{population of Sweden}}{\text{population of Fiji}}$$

$$k = \frac{9.9 \times 10^6}{9.1 \times 10^5} = 10.879 \dots$$

5. Convert standard form

To the nearest whole number,

$$k = 11$$

FINAL ANSWER(a) 1.318×10^9 , (b) 11

PAST PAPER QUESTION**Q#: 9****Marks: 5****TOPIC: Factorising**

May 2021 P1H - May 2021 | P1H

9 (a) Factorise fully $25a^4c^7d + 45a^9c^3h$

.....
(2)

(b) Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

$x =$
(3)

(Total for Question 9 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 5****TOPIC: Factorising**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$25a^4c^7d + 45a^9c^3h = 5a^4c^3(5c^4d + 9a^5h)$$

2. (b)

(b)

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

$$4x^2 + 20x + 25 = 4x^2 + 4x - 3$$

$$16x = -28$$

$$x = -\frac{7}{4}$$

FINAL ANSWER(a) $5a^4c^3(5c^4d + 9a^5h)$, (b) $x = -\frac{7}{4}$

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Statistics Toolkit**

May 2021 P1H - May 2021 | P1H

10 Jethro has sat 5 tests.

Each test was marked out of 100 and Jethro's mean mark for the 5 tests is 74

Jethro has to sit one more test that is also to be marked out of 100

Jethro wants his mean mark for all 6 tests to be at least 77

Work out the least mark that Jethro needs to get for the last test.

(Total for Question 10 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 10** **Marks: 3**TOPIC: **Statistics Toolkit**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The total for the first**

The total for the first 5 tests is

$$5 \times 74 = 370$$

2. Calculate value

For an average of at least 77 after 6 tests, the total must be at least

$$6 \times 77 = 462$$

3. Calculate value

So the last test score must be at least

$$462 - 370 = 92$$

FINAL ANSWER

at least 92.

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Algebraic Roots & Indices**

May 2021 P1H - May 2021 | P1H

11 $\sqrt{2} \times 16 = 2^x$

- (a) Find the value of x .
Show your working clearly.

$$x = \dots\dots\dots (2)$$

$$\frac{(11^{-6})^5}{11^4} = 11^n$$

- (b) Find the value of n .
Show your working clearly.

$$n = \dots\dots\dots (2)$$

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 4****TOPIC: Algebraic Roots & Indices**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\sqrt{2} \times 16 = 2^x$$

2. Write both factors as powers

Write both factors as powers of 2:

$$\sqrt{2} = 2^{1/2}, \quad 16 = 2^4$$

3. Use index laws

So

$$2^{1/2} \times 2^4 = 2^{9/2}$$

4. Use index laws

Therefore

$$x = \frac{9}{2}$$

5. Use index laws

For part (b),

$$\frac{(11^{-6})^5}{11^4} = \frac{11^{-30}}{11^4} = 11^{-30-4} = 11^{-34}$$

6. Use index laws

So

$$n = -34$$

FINAL ANSWER(a) $x = \frac{9}{2}$, (b) $n = -34$.

PAST PAPER QUESTION**Q#: 12****Marks: 2****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

- 12** The diagram shows a sector of a circle with radius 7 cm.

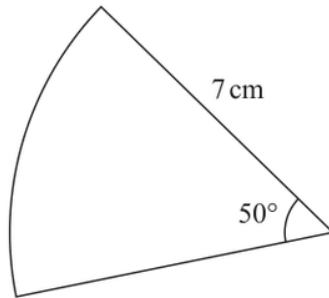


Diagram **NOT**
accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

(Total for Question 12 is 2 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 2

TOPIC: **Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$\begin{aligned}\text{arc length} &= \frac{50}{360} \times 2\pi(7) \\ &= 6.108\dots\end{aligned}$$

FINAL ANSWER

6.1 cm.

EA

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Expanding Brackets**

May 2021 P1H - May 2021 | P1H

- 13** Expand and simplify $4x(3x + 1)(2x - 3)$
Show your working clearly.

(Total for Question 13 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Expanding Brackets**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Expand brackets

$$4x(3x + 1)(2x - 3) = 4x(6x^2 - 7x - 3)$$

$$= 24x^3 - 28x^2 - 12x$$

FINAL ANSWER

$$24x^3 - 28x^2 - 12x$$

EA

PAST PAPER QUESTION**Q#: 14****Marks: 2****TOPIC: Statistics Toolkit**

May 2021 P1H - May 2021 | P1H

14 Here is the number of goals that Henri's team scored one summer in each water polo match.

5 8 9 11 13 13 14 15 16 17 20

Find the interquartile range of the numbers of goals.
Show your working clearly.

(Total for Question 14 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 2

TOPIC: **Statistics Toolkit**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The goals in order are**

The goals in order are

5, 8, 9, 11, 13, 13, 14, 15, 16, 17, 20

$$Q_1 = 9, \quad Q_3 = 16$$

$$\text{IQR} = 16 - 9 = 7$$

FINAL ANSWER

7.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Circle Theorems**

May 2021 P1H - May 2021 | P1H

- 15** P , Q and R are points on a circle, centre O .
 TRV is the tangent to the circle at R .

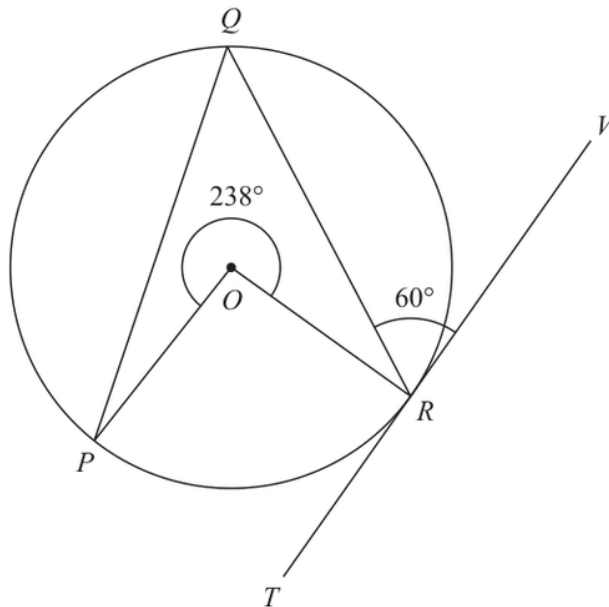


Diagram **NOT**
 accurately drawn

Reflex angle $POR = 238^\circ$

Angle $QRV = 60^\circ$

Calculate the size of angle OPQ .

Give a reason for each stage of your working.

o

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 4****TOPIC: Circle Theorems**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

The reflex angle POR is 238° , so the smaller angle is

$$360^\circ - 238^\circ = 122^\circ$$

2. Use trigonometry

Since $OP = OR$, triangle OPR is isosceles:

$$\angle OPR = \frac{180^\circ - 122^\circ}{2} = 29^\circ$$

3. Use trigonometry

By the alternate segment theorem,

$$\angle QPR = 60^\circ$$

4. Use trigonometry

Therefore

$$\angle QPO = 60^\circ - 29^\circ = 31^\circ$$

FINAL ANSWER

31° .

PAST PAPER QUESTION**Q#: 16****Marks: 2****TOPIC: Algebraic Proof**

May 2021 P1H - May 2021 | P1H

16 Use algebra to show that the recurring decimal $0.28\dot{1}\dot{3} = \frac{557}{1980}$

(Total for Question 16 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 16****Marks: 2****TOPIC: Algebraic Proof**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let**

Let

$$x = 0.28131313 \dots$$

2. Calculate value

Then

$$100x = 28.131313 \dots$$

3. Calculate value

and

$$10000x = 2813.131313 \dots$$

4. Subtract

Subtract:

$$10000x - 100x = 2813.131313 \dots - 28.131313 \dots$$

$$9900x = 2785$$

$$x = \frac{2785}{9900} = \frac{557}{1980}$$

FINAL ANSWER

$$0.28131313 \dots = \frac{557}{1980}$$

PAST PAPER QUESTION**Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

May 2021 P1H - May 2021 | P1H

- 17** Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

(Total for Question 17 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the three consecutive even**

Let the three consecutive even numbers be

$$2n - 2, \quad 2n, \quad 2n + 2$$

2. The middle number isThe middle number is $2n$.**3. The difference between the square**

The difference between the square of the largest and the square of the smallest is

$$\begin{aligned} & (2n + 2)^2 - (2n - 2)^2 \\ &= (4n^2 + 8n + 4) - (4n^2 - 8n + 4) \\ &= 16n \end{aligned}$$

4. Calculate valueSince $8 \times \text{middle number} = 8(2n) = 16n$, the result is 8 times the middle number.**FINAL ANSWER**

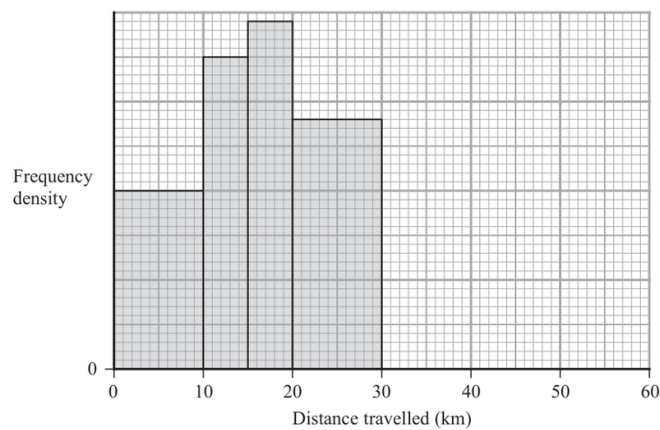
The difference is always 8 times the middle number.

PAST PAPER QUESTION**Q#: 18****Marks: 5****TOPIC: Histograms**

May 2021 P1H - May 2021 | P1H

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 5****TOPIC: Histograms**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use histogram**

The $0 \leq d < 10$ class has frequency 40, so its density is

$$\frac{40}{10} = 4$$

2. Use histogram

From the histogram, the missing frequencies are

$$10 \leq d < 15 : 35, \quad 15 \leq d < 20 : 40, \quad 20 \leq d < 30 : 60$$

3. Together with the table

Together with the table:

$$40 + 35 + 40 + 60 + 30 = 205$$

4. Use histogram

For $d > 25$:

$$5(6) + 30 = 60$$

$$P(d > 25) = \frac{60}{205} = \frac{12}{41}$$

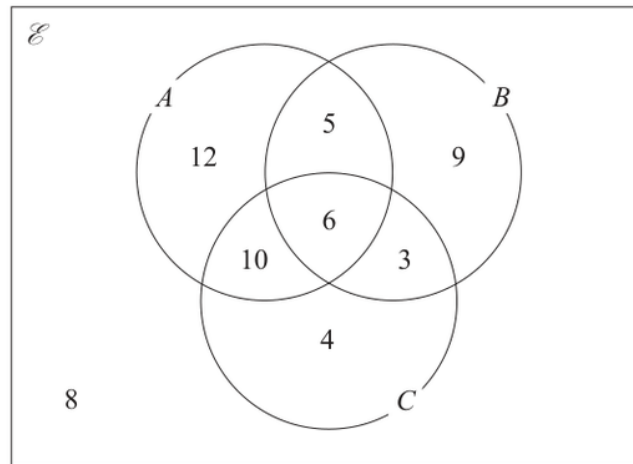
FINAL ANSWER

completed frequencies 35, 40, 60, missing histogram density 1, and probability $\frac{12}{41}$.

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

May 2021 P1H - May 2021 | P1H

19 The Venn diagram shows a universal set, $\mathcal{E}$ and sets A , B and C .



12, 5, 9, 10, 6, 3, 4 and 8 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....
(1)

(ii) $n(A' \cap B')$

.....
(1)

(iii) $n([A \cap B] \cup C)$

.....
(1)

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. From the Venn diagram**

From the Venn diagram:

2. only is only is only

- A only is 12
- $A \cap B$ only is 5
- B only is 9
- $A \cap C$ only is 10
- all three is 6
- $B \cap C$ only is 3
- C only is 4
- outside is 8

3. Add every region insideFor $A \cup B$, add every region inside A or B :

$$12 + 5 + 9 + 10 + 6 + 3 = 45$$

4. Add the regions outside bothFor $A' \cap B'$, add the regions outside both A and B :

$$4 + 8 = 12$$

5. Add and allFor $(A \cap B) \cup C$, add $A \cap B$ and all of C :

$$5 + 6 + 10 + 3 + 4 = 28$$

FINAL ANSWER

(i) 45, (ii) 12, (iii) 28

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May 2021 P1H - May 2021 | P1H

20 $P = \frac{t - w}{y}$

 $t = 9.7$ correct to 1 decimal place $w = 5.9$ correct to 1 decimal place $y = 3$ correct to 1 significant figure

Calculate the upper bound for the value of P .
Show your working clearly.

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Estimate the value**

$$P = \frac{t - w}{y}$$

2. Find upper bound

For the upper bound, make the numerator large and the denominator small.

$$t < 9.75, \quad w \geq 5.85, \quad y \geq 2.5$$

$$P_{\text{upper}} = \frac{9.75 - 5.85}{2.5} = 1.56$$

FINAL ANSWER

1.56

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May 2021 P1H - May 2021 | P1H

20 $P = \frac{t - w}{y}$

 $t = 9.7$ correct to 1 decimal place $w = 5.9$ correct to 1 decimal place $y = 3$ correct to 1 significant figure

Calculate the upper bound for the value of P .
Show your working clearly.

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Estimate the value**

$$P = \frac{t - w}{y}$$

2. Find upper bound

For the upper bound of P , make the numerator as large as possible and the denominator as small as possible.

$$t = 9.75, \quad w = 5.85, \quad y = 2.5$$

$$P = \frac{9.75 - 5.85}{2.5}$$

$$P = \frac{3.90}{2.5} = 1.56$$

FINAL ANSWER

1.56

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

May 2021 P1H - May 2021 | P1H

21 Given that $x = \frac{5}{9y+5}$ and that $y = \frac{5}{5a-2}$

find an expression for x in terms of a .

Give your expression as a single fraction in its simplest form.

(Total for Question 21 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Given**

Given

$$x = \frac{5}{9y + 5}$$

2. Use trigonometry

and

$$y = \frac{5}{5a - 2}$$

3. Use cosine ruleSubstitute $y = \frac{5}{5a-2}$ into the expression for x :

$$x = \frac{5}{9\left(\frac{5}{5a-2}\right) + 5}$$

$$x = \frac{5}{\frac{45}{5a-2} + 5}$$

4. Write the denominator as a

Write the denominator as a single fraction:

$$\frac{45}{5a-2} + 5 = \frac{45 + 5(5a-2)}{5a-2}$$

$$= \frac{45 + 25a - 10}{5a-2}$$

$$= \frac{25a + 35}{5a-2}$$

5. Use trigonometry

So

$$x = 5 \div \frac{25a + 35}{5a - 2}$$

$$x = 5 \times \frac{5a - 2}{25a + 35}$$

$$x = \frac{5(5a - 2)}{5(5a + 7)}$$

$$x = \frac{5a - 2}{5a + 7}$$

FINAL ANSWER

$$\frac{5a - 2}{5a + 7}$$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

May 2021 P1H - May 2021 | P1H

21 Given that $x = \frac{5}{9y+5}$ and that $y = \frac{5}{5a-2}$

find an expression for x in terms of a .

Give your expression as a single fraction in its simplest form.

(Total for Question 21 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Given**

Given

$$x = \frac{5}{9y + 5}$$

2. Use trigonometry

and

$$y = \frac{5}{5a - 2}$$

3. Use cosine ruleSubstitute $y = \frac{5}{5a-2}$ into the expression for x :

$$x = \frac{5}{9\left(\frac{5}{5a-2}\right) + 5}$$

$$x = \frac{5}{\frac{45}{5a-2} + 5}$$

4. Write the denominator as a

Write the denominator as a single fraction:

$$\frac{45}{5a-2} + 5 = \frac{45 + 5(5a-2)}{5a-2}$$

$$= \frac{45 + 25a - 10}{5a-2}$$

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So

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$$x = 5 \times \frac{5a - 2}{25a + 35}$$

$$x = \frac{5(5a - 2)}{5(5a + 7)}$$

$$x = \frac{5a - 2}{5a + 7}$$

FINAL ANSWER

$$\frac{5a - 2}{5a + 7}$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

May 2021 P1H - May 2021 | P1H

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

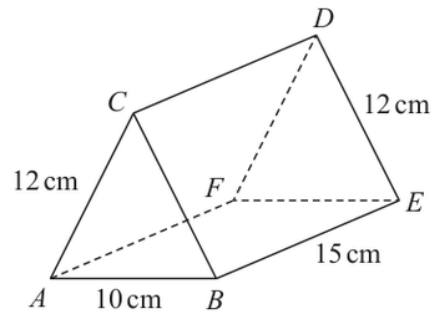


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.

Give your answer correct to 3 significant figures.

°

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

The triangular end is isosceles, with

$$AC = BC = 12, \quad AB = 10$$

2. The perpendicular height of theThe perpendicular height of the triangle is found by halving AB :

$$AM = 5$$

$$CM^2 = 12^2 - 5^2 = 144 - 25 = 119$$

$$CM = \sqrt{119}$$

3. The projection of on theThe projection of AD on the horizontal base goes from A to the midpoint of FE . Its length is

$$\sqrt{15^2 + 5^2} = \sqrt{250}$$

4. Use trigonometrySo, for the angle θ between AD and the base,

$$\tan \theta = \frac{\sqrt{119}}{\sqrt{250}}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{119}}{\sqrt{250}} \right) = 34.602 \dots$$

FINAL ANSWER**34.6°**

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

May 2021 P1H - May 2021 | P1H

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

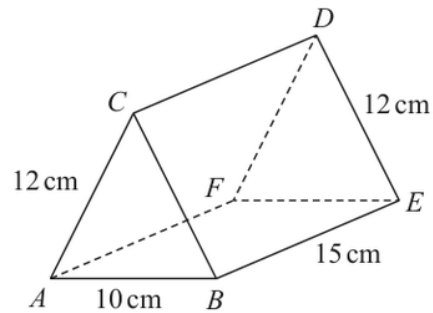


Diagram **NOT**
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Calculate the size of the angle between AD and the base $ABEF$.

Give your answer correct to 3 significant figures.

°

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

May 2021 P1H - May 2021 | P1H

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The triangular end is isosceles, with

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FINAL ANSWER**34.6°**

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Sequences**

May 2021 P1H - May 2021 | P1H

23 The sum of the first N terms of an arithmetic series, S , is 292

The 2nd term of S is 8.5

The 5th term of S is 13

Find the value of N .

Show clear algebraic working.

$N =$

(Total for Question 23 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Sequences**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The 2nd term is**

The 2nd term is

$$a + d = 8.5$$

2. The 5th term is

The 5th term is

$$a + 4d = 13$$

3. Subtract

Subtract:

$$3d = 4.5$$

$$d = 1.5$$

$$a = 8.5 - 1.5 = 7$$

4. Use

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$292 = \frac{N}{2}(14 + 1.5(N - 1))$$

$$292 = \frac{N}{2}(1.5N + 12.5)$$

$$3N^2 + 25N - 1168 = 0$$

$$N = \frac{-25 + 121}{6} = 16$$

FINAL ANSWER

$$N = 16$$

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Sequences**

May 2021 P1H - May 2021 | P1H

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Find the value of N .

Show clear algebraic working.

$N =$

(Total for Question 23 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Sequences**

May 2021 P1H - May 2021 | P1H

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FINAL ANSWER

$$N = 16$$

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Functions**

May 2021 P1H - May 2021 | P1H

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$
(2)(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots$$

(4)

(Total for Question 24 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Functions**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$g(2) = 7(2) - 6 = 8$$

$$fg(2) = f(8)$$

$$f(8) = 5(8)^2 - 10(8) + 7$$

$$f(8) = 247$$

2. (b)

(b)

$$f(x) = 5x^2 - 10x + 7$$

$$f(x) = 5(x - 1)^2 + 2$$

3. Let

Let

$$y = 5(x - 1)^2 + 2$$

$$(x - 1)^2 = \frac{y - 2}{5}$$

4. Find inverse functionSince the domain is $x \geq 1$, take the positive square root:

$$x = 1 + \sqrt{\frac{y - 2}{5}}$$

FINAL ANSWER

$$(a) 247, (b) f^{-1}(x) = 1 + \sqrt{\frac{x - 2}{5}}$$

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Functions**

May 2021 P1H - May 2021 | P1H

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(4)

(Total for Question 24 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Functions**

May 2021 P1H - May 2021 | P1H

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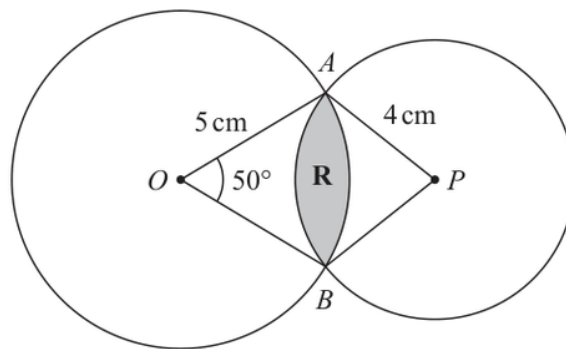
FINAL ANSWER

$$(a) 247, (b) f^{-1}(x) = 1 + \sqrt{\frac{x - 2}{5}}$$

PAST PAPER QUESTION**Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

- 25** The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

Diagram **NOT**
accurately drawn

One of the circles has centre O and radius 5 cm .
 The other circle has centre P and radius 4 cm .
 Angle $AOB = 50^\circ$

Calculate the area of region **R**.
 Give your answer correct to 3 significant figures.

..... cm^2 **(Total for Question 25 is 6 marks)**

PAST PAPER SOLUTION**Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**For the circle with centre O :

$$\angle AOB = 50^\circ, \quad OA = OB = 5$$

2. The half chord length is

The half chord length is

$$\frac{AB}{2} = 5 \sin 25^\circ$$

3. Use trigonometryFor the circle with centre P , let

$$\angle APB = \theta$$

4. Use trigonometry

The radius is 4, so

$$4 \sin \left(\frac{\theta}{2} \right) = 5 \sin 25^\circ$$

$$\sin \left(\frac{\theta}{2} \right) = \frac{5 \sin 25^\circ}{4}$$

$$\theta = 63.7777 \dots^\circ$$

5. The common region is made

The common region is made from two minor segments.

6. Use trigonometryFor the circle with centre O :

$$\text{segment area} = \frac{50}{360} \pi (5)^2 - \frac{1}{2} (5)^2 \sin 50^\circ = 1.3328 \dots$$

7. Use trigonometryFor the circle with centre P :

$$\begin{aligned} \text{segment area} &= \frac{\theta}{360} \pi (4)^2 - \frac{1}{2} (4)^2 \sin \theta \\ &= 1.7284 \dots \end{aligned}$$

8. Total area of region

Total area of region R :

$$1.3328\dots + 1.7284\dots = 3.0611\dots$$

FINAL ANSWER

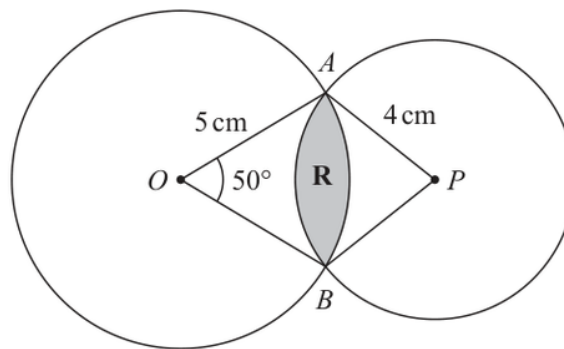
$$3.06 \text{ cm}^2$$

EA

PAST PAPER QUESTION**Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

- 25** The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

Diagram **NOT**
accurately drawn

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 Give your answer correct to 3 significant figures.

..... cm^2 **(Total for Question 25 is 6 marks)**

PAST PAPER SOLUTION**Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**For the circle with centre O :

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Total area of region R :

$$1.3328\dots + 1.7284\dots = 3.0611\dots$$

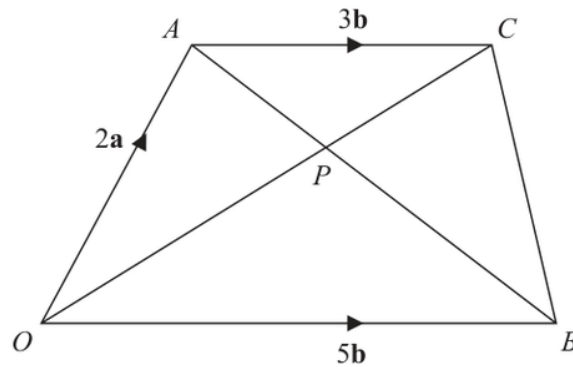
FINAL ANSWER

$$3.06 \text{ cm}^2$$

EA

PAST PAPER QUESTION**Q#: 26****Marks: 5**TOPIC: **Vectors**

May 2021 P1H - May 2021 | P1H

26 $OACB$ is a trapezium.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

The diagonals, OC and AB , of the trapezium intersect at the point P .Find and simplify an expression, in terms of $\mathbf{a}$ and $\mathbf{b}$, for $\vec{OP}$
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26

Marks: 5

TOPIC: **Vectors**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use vectors

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 5\mathbf{b}, \quad \overrightarrow{AC} = 3\mathbf{b}$$

2. Use vectors

So

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = 2\mathbf{a} + 3\mathbf{b}$$

3. Use vectorsSince P lies on OC , let

$$\overrightarrow{OP} = t(2\mathbf{a} + 3\mathbf{b})$$

4. Use vectorsSince P also lies on AB , let

$$\overrightarrow{OP} = \overrightarrow{OA} + s\overrightarrow{AB}$$

5. Use vectors

Now

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 5\mathbf{b} - 2\mathbf{a}$$

6. Use vectors

So

$$\overrightarrow{OP} = 2\mathbf{a} + s(5\mathbf{b} - 2\mathbf{a})$$

$$\overrightarrow{OP} = (2 - 2s)\mathbf{a} + 5s\mathbf{b}$$

7. Use vectors

Compare this with

$$t(2\mathbf{a} + 3\mathbf{b}) = 2t\mathbf{a} + 3t\mathbf{b}$$

8. Equating coefficients

Equating coefficients:

$$2t = 2 - 2s$$

$$3t = 5s$$

9. Use vectors

From $t = 1 - s$,

$$3(1 - s) = 5s$$

$$3 = 8s$$

$$s = \frac{3}{8}$$

10. Use vectors

So

$$t = 1 - \frac{3}{8} = \frac{5}{8}$$

11. Use vectors

Therefore

$$\overrightarrow{OP} = \frac{5}{8}(2\mathbf{a} + 3\mathbf{b})$$

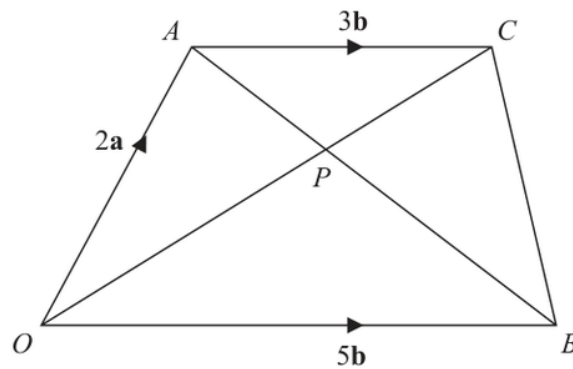
$$\overrightarrow{OP} = \frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

FINAL ANSWER

$$\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

PAST PAPER QUESTION**Q#: 26****Marks: 5****TOPIC: Vectors**

May 2021 P1H - May 2021 | P1H

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Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION**Q#: 26****Marks: 5**TOPIC: **Vectors**

May 2021 P1H - May 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use vectors**

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 5\mathbf{b}, \quad \overrightarrow{AC} = 3\mathbf{b}$$

2. Use vectors

So

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = 2\mathbf{a} + 3\mathbf{b}$$

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Now

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 5\mathbf{b} - 2\mathbf{a}$$

6. Use vectors

So

$$\overrightarrow{OP} = 2\mathbf{a} + s(5\mathbf{b} - 2\mathbf{a})$$

$$\overrightarrow{OP} = (2 - 2s)\mathbf{a} + 5s\mathbf{b}$$

7. Use vectors

Compare this with

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10. Use vectors

So

$$t = 1 - \frac{3}{8} = \frac{5}{8}$$

11. Use vectors

Therefore

$$\overrightarrow{OP} = \frac{5}{8}(2\mathbf{a} + 3\mathbf{b})$$

$$\overrightarrow{OP} = \frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

FINAL ANSWER

$$\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$